IEEE CLIMATECHAIN GLOBAL HACKATHON 2026 · TRACK: CLIMATE DATA & ENVIRONMENTAL MONITORING

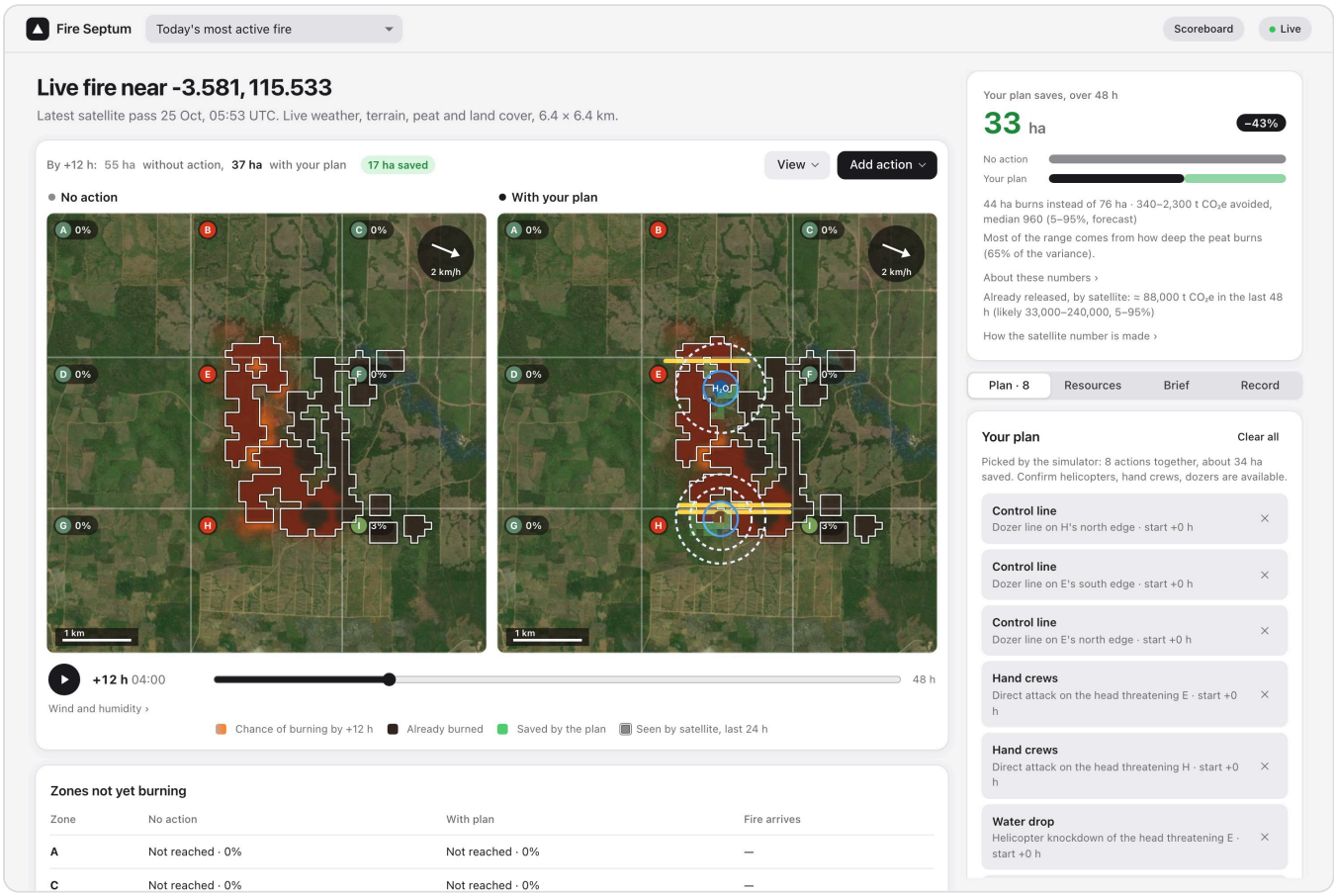
Fire Septum

Peat-fire forecasts you can hold to account. We predict where Indonesia's fires spread next, seal each forecast on Bitcoin before it happens, then score it in public against NASA satellites.

LIVE APP
fire-septum.vercel.app

CODE
github.com/aliefauzan/fire-septum

DEMO VIDEO
youtu.be/dcFxrvgIWsQ



A real fire in Kalimantan, seen by NASA satellites on 25 Oct 2026. Left: what the simulation expects with no action. Right: with the plan it picked (control lines, a water drop, hand crews). Everything on this page is live data.

Sekat api is Indonesian for "fire barrier". This pack is for judges: the problem, what we built, what we measured, and how you can check it yourself without trusting us. The last page is a real commander brief exported by the app.

The problem: nobody keeps score of fire forecasts

In 2015, Indonesia's peat fires released more CO_2 per day than the whole US economy (WRI, from GFED estimates). Drained tropical peat releases about **800 t CO_2e per hectare** when it burns (IPCC Tier 1), and it can smoulder underground for weeks.

Every dry season, BPBD and Manggala Agni commanders decide where to send crews, dozers and helicopters. The data they need is scattered across NASA satellites, weather services, terrain, peat maps, canals and roads. And no one keeps score of the forecasts they act on:

- You cannot prove what a model said **before** the fire spread.
- A model can always be tuned afterwards to look good on past fires.
- So no one knows which forecasts deserve trust, and decisions leave no record anyone can check.

This is the track's problem: environmental data that is **fragmented and hard to verify**.

Our answer: one forecast, made accountable

1 Forecast

A gradient-boosted model gives the chance of new fire tomorrow in every 400 m block (trained on 2015 and 2019, tested on fires it never saw). A 40-run simulation shows how the fire spreads over 24–48 h, with the uncertainty visible.

2 Plan and count the carbon

Control lines, helicopter drops, hand crews and peat injection are simulated at several start times and ranked by hectares saved, using only resources the commander says exist. CO_2e avoided is shown as a range.

3 Seal before the outcome

Every forecast and decision is hashed (SHA-256), signed (Ed25519) and timestamped on Bitcoin with OpenTimestamps. Daily Merkle roots can be attested on Base (EAS). Re-running a record gives bit-identical numbers.

4 Score in public

After 24 h, each sealed forecast is scored against NASA VIIRS detections and stays on a public scoreboard, good or bad, next to two fixed reference models. Anyone can submit a model in the open format.

It also watches fire in and around **REDD+ peat carbon projects** and seals the satellite evidence, and it exports a **one-page PDF brief** for commanders in Bahasa Indonesia or English.

> **Why a blockchain, and only this much.** The one trust question is "was this said before the fire spread?". A Bitcoin timestamp answers exactly that, with no wallet, fee or token. Fire data never goes on chain; a smart contract could not make a simulation more correct. Re-running does that.

What we measured, not claimed

The train / validation / test split was fixed by a stated rule and **timestamped on Bitcoin before any model was trained** (`data/validation/events.json.ots`): 1,492 fire events plus 2,700 training boxes.

AUC 0.79
next-day forecaster on held-out 2023 Sumatra fires (184 boxes)

4 engines
identical output hash: V8 (Node, Chromium), JavaScriptCore, SpiderMonkey

191 tests
run in CI on every push, with the cross-engine check

Next-day forecaster vs simple references, same rows (test set: 2023 Sumatra)

MODEL	BRIER SKILL [95% CI]	CSI	AUC
Fire Septum forecaster (calibrated)	**+0.036 [+0.026, +0.045]**	**0.08**	**0.79**
Distance to fire	+0.022 [+0.016, +0.028]	0.06	0.76
Persistence	+0.020 [+0.015, +0.026]	0.05	0.72
Climatology	0.000	0.02	0.50

Spread simulation hindcast, 12 real fires, 8–18 Oct 2026

REGION	EVENTS	NEW-FIRE BLOCKS	CSI, MODEL	CSI, DISTANCE ONLY
Pooled	12	114	0.23	0.18
Kalimantan	6	90	0.35	0.25
Sumatra (held out)	6	24	0.12	0.11

Pre-registered forecasts: sealed before the outcome, scored after

MODEL	BOXES	CSI	POD	FAR
Distance (reference)	6	0.18	0.72	0.80
Fire Septum simulation	**6**	**0.13**	**0.68**	**0.86**
Climatology (reference)	6	0.00	0.00	0.00

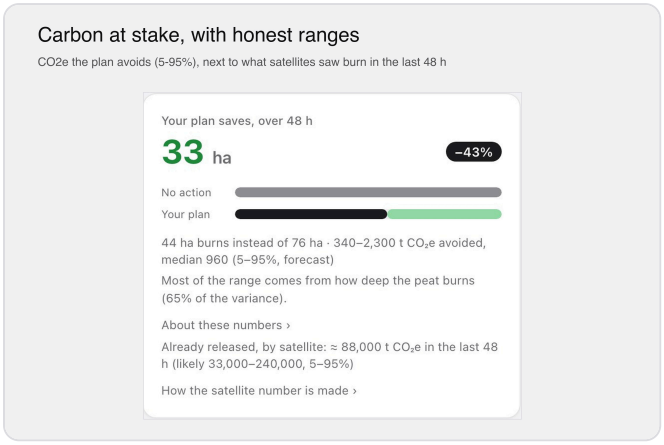
> On the register our simulation currently scores **below** the distance reference. We publish that. It is the point of the register: a sealed forecast cannot be tuned after the fact, so the scoreboard tells the truth.

Carbon cross-checks

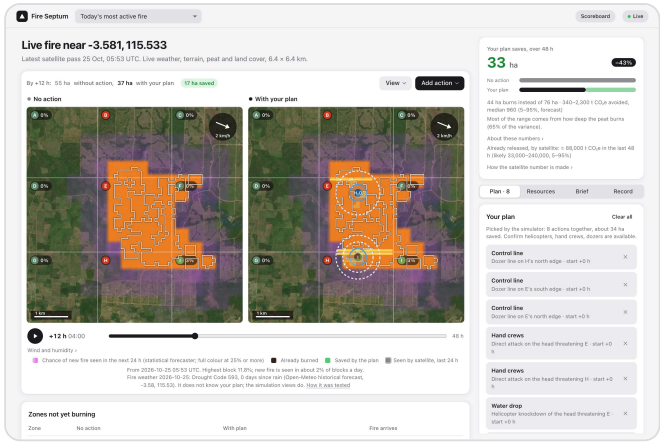
- Satellite (FRP) CO_2 vs GFED4.1s in the five most-burned peat cells: 3.04× (2015), 1.21× (2019), 1.02× (2023).
- Burned area measured afterwards from Sentinel-2 (dNBR) and Sentinel-1 (SAR) for 9 of the October 2026 fires: 81–100% lies inside the FIRMS footprints.
- Every emission number is a Monte Carlo 5–95% range; peat burn depth comes from each cell's fire history (Konecny et al. 2016).

Sources: `docs/validation-scale.json`, `docs/hindcast-rerun-2026-10-24-barriers.json`, `data/register/`, `docs/determinism.json`. These numbers are generated from the data files and checked in CI.

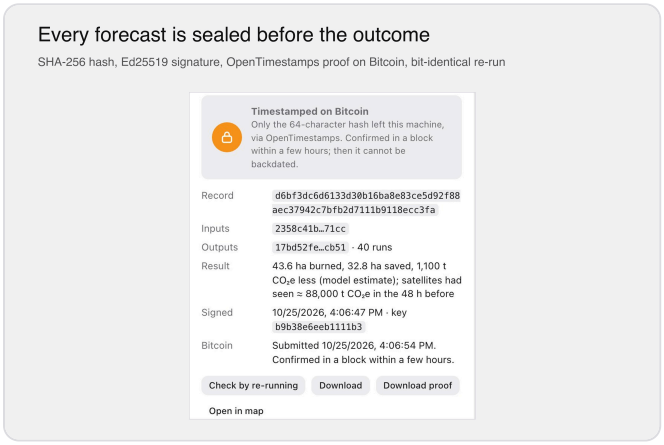
Features that matter for the climate



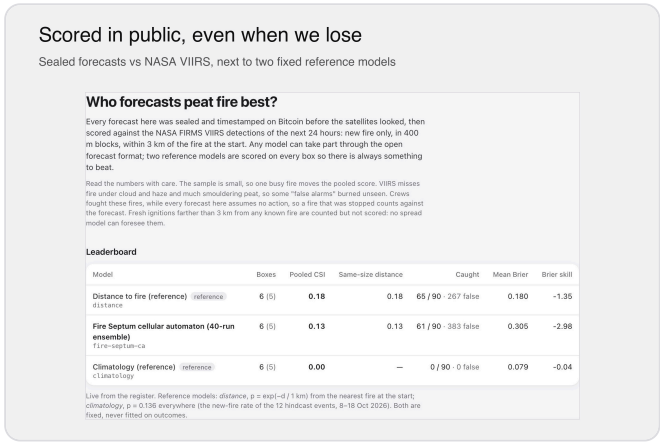
Carbon at stake: CO_2e the plan avoids as a 5–95% range, and what satellites saw burn in the last 48 h.



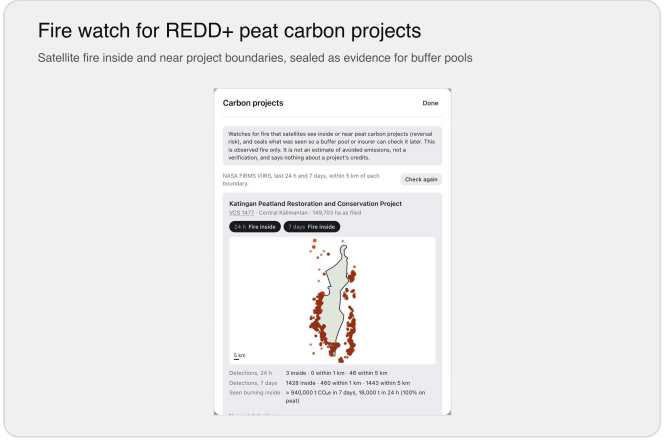
Tomorrow view: chance of new fire in every 400 m block in the next 24 h.



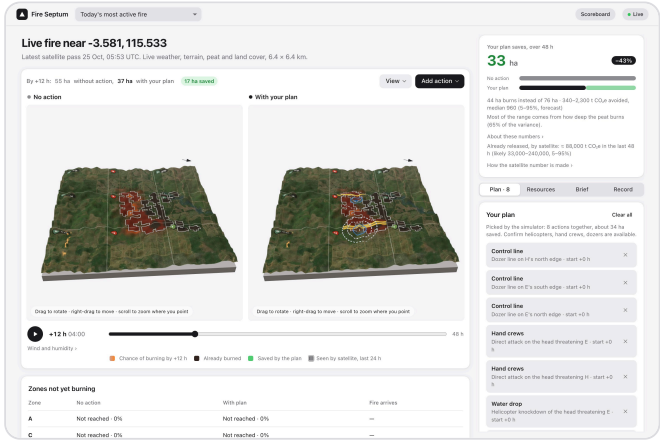
Sealed on Bitcoin before the outcome. Change one byte of input and the hash changes.



Public scoreboard against NASA VIIRS, next to two fixed reference models.



Fire watch for REDD+ peat carbon projects: reversal-risk evidence for buffer pools and insurers.



The same forecast on 3D terrain from real elevation and satellite imagery.

Check it yourself, without trusting us

In the browser (2 minutes)

1. Open **fire-septum.vercel.app**. It loads today's most active fire in Kalimantan from NASA FIRMS.
2. Open the **Record** tab and press **Seal and sign**, then **Timestamp on Bitcoin**.
3. Press **Check by re-running**: the app re-runs all 40 simulations and recomputes every hash.
4. Press **Download proof** to get the receipt and a standard `.ots` file.
5. Open **Scoreboard** to see every pre-registered forecast and its score.

With the reference OpenTimestamps client (no Fire Septum code involved)

```
pip install opentimestamps-client
ots upgrade fireseptum-<hash>.receipt.json.ots   # once the Bitcoin block is confirmed
ots verify  fireseptum-<hash>.receipt.json.ots
```

From the code

```
git clone https://github.com/aliefauzan/fire-septum && cd fire-septum
npm install && npm test              # 191 tests
node scripts/determinism.mjs         # same output hash in Node, Chromium, Firefox, WebKit
node scripts/export-proofs.mjs       # every register receipt + .ots proof, for ots verify
npx tsx scripts/gen-docs.ts --check  # every number in the docs matches the data files
```

Questions a skeptical judge may ask

Isn't OpenTimestamps trivial?	The cryptography is simple on purpose. What is new is what gets sealed: a full reproducible forecast (inputs and every simulated cell), sealed before the outcome exists and scored after. That is pre-registration for climate models.
Couldn't you pick the fires you predict well?	No. The register picks the top 3 FIRMS clusters per island by a fixed rule at issue time, and keeps every forecast.
Couldn't you edit a forecast later?	Its hash is in a Bitcoin block. Changing one input, rate or simulated cell changes the hash, and *Check by re-running* fails.
Is the model any good?	The next-day forecaster beats persistence and distance on held-out fires. The spread simulation is only slightly better than distance in hindcast and below it on the register so far. We show both.
Are the CO_2 numbers credits?	No. They are IPCC 2013 Wetlands factors with per-cell burn depth, shown as ranges and cross-checked against GFED and Sentinel burned area. Not credits, not claimable avoided emissions.
What about AI?	The core is a physical-statistical model, not a black box. An optional LLM only rewrites the simulated numbers into a brief, and it is off by default.
Who has used it?	Nobody operational yet. Next step is testing the planning view and PDF brief with one Daops Manggala Agni unit.

Honest limits

- Peat fires smoulder and creep, so a flame-front cellular automaton fits them poorly. Its skill over distance-only is small and its raw probabilities are not calibrated. This is why we added the separate statistical forecaster.
- Peat burn depth is inferred from satellite fire history, not measured in the field.
- The satellite (FRP) emission number runs 1–3× GFED and can disagree with the forecast number by orders of magnitude. Both are shown with their method.
- OpenStreetMap canals are patchy, so the drainage effect is an assumption.
- SiPongi (Kemenhut) data needs written permission, so that adapter is off.
- It has not yet been tested with BPBD or Manggala Agni staff.

Who it is for

- **BPBD and Manggala Agni commanders:** where fire is likely tomorrow and what acting would change, with uncertainty visible and no invented resources.
- **Provincial governments:** when to request aircraft, on a record that shows what was known at the time.
- **Researchers and agencies:** a public, pre-registered scoreboard where any fire model can be compared.
- **Carbon-project verifiers, buffer pools and auditors:** proof of what was forecast, decided and observed, and when.

What's next

- Run the register through the 2026 dry season and invite other models onto the scoreboard.
- Add peat-moisture data (SMAP / SiPALAGA) to the forecaster.
- Test the planning view and the PDF brief with one Daops unit.

Built with

TypeScript, React, Vite, three.js, Web Workers · Python, scikit-learn · NASA FIRMS, ERA5 / Open-Meteo, AWS Terrain Tiles, Global Forest Watch peat, ESA WorldCover, OpenStreetMap, Sentinel-1/2, BMKG · OpenTimestamps (Bitcoin), Ethereum Attestation Service (Base), Ed25519, RFC 6962 Merkle trees · GitHub Actions, Vercel, Vitest, Playwright.

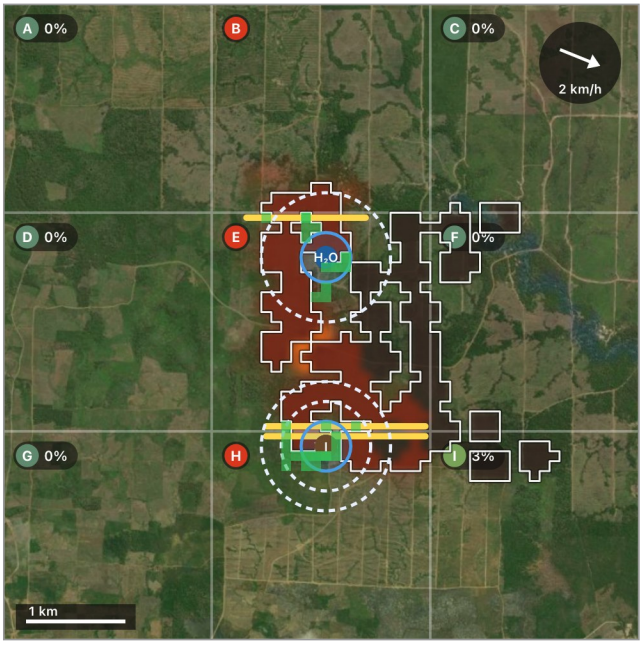
Research prototype using a simplified fire model. Not firefighting guidance: follow BPBD, Manggala Agni and incident command. Emission numbers are model or satellite estimates, not measurements and not carbon credits. Next page: a real situation brief exported by the app on 25 Oct 2026.

Live fire near -3.581, 115.533

Forecast start: 16:00 Location: -3.5812, 115.5329

Data: Real data. Weather: Open-Meteo hourly forecast (real), fetched 10/25/2026, 4:03:33 PM. Fires: NASA FIRMS VIIRS_SNPP_NRT + VIIRS_NOAA20_NRT + VIIRS_NOAA21_NRT, last 48 h (41 detections in area).

Fuel and peat: Peat soil: Global Forest Watch peatland map (real), 8% of cells. Land cover: ESA WorldCover 2021 (real), 100% of cells. Barriers: OpenStreetMap snapshot of 2026-10-23 (Geofabrik extract, 16 features): canals, rivers, major roads, water and built-up areas non-burnable on 1% of cells (49); ditches, drains and streams (fuel-break strength 0.5) and minor roads and tracks (0.3) slow spread on 11% (470 cells). Drained peat: spread hazard on peat soil × (1 + 0.5·exp(-d/100 m)), d = distance to the nearest canal, ditch or drain. Fire history: NASA FIRMS MODIS and VIIRS detections 2001-2025 (real), years before 2026 only; of 332 peat cells 8% had no earlier fire year, 6% one, 14% two, 72% three or more. Terrain: AWS Terrain Tiles (real).



With your plan: chance of burning by +12 h. Dark = already burned; outlined = seen by satellite, last 24 h.

Burns in 48 h if nothing is done

76 ha

likely 52-105 ha

With the current plan (8 actions)

44 ha

likely 28-66 ha

Peat emissions, forecast if nothing is done

710-4,100 t CO2e

median 1,800, 5-95% over fire spread, burn depth and emission factors

Already released, seen by satellite (last 48 h)

~88,000 t CO2e

VIIRS FRP, 41 detections, 18% peat; a floor for smouldering peat

Resources entered

Helicopters unknown · Hand crews unknown · Dozers unknown · Seeding aircraft unknown

(not entered)

ZONE RISK

Zone	Status	Runs ignited	With plan	Fire arrives (p10-p90)
A	Not reached	0%	0%	-
B	**Burning**	**-**	**-**	**-**
C	Not reached	0%	0%	-
D	Not reached	0%	0%	-
E	**Burning**	**-**	**-**	**-**
F	Not reached	0%	0%	-
G	Not reached	0%	0%	-
H	**Burning**	**-**	**-**	**-**
I	Lower risk	3%	3%	+6.5 h - +6.5 h

Share of 40 simulated runs in which the zone catches fire within 48 h. No forecast is a guarantee.

TOP RECOMMENDATIONS (SIMULATED, RANKED BY AREA SAVED)

1. Helicopter knockdown of the head threatening B (-2 ha; resources not confirmed)

Where fire head nearest to Zone B, 3.2 km E / 1.8 km S of the NW corner (-3.5690, 115.5334), radius 0.2 km.

What Drop 26 helicopter buckets on the active head (~13 ha) to knock down flames and wet the fuel just ahead of it. Estimated flame length at the head ~ 0.3 m in plantation (oil palm / acacia): hand crews can work the head or flanks (NWCG IRPG (PMS 461, 2025); Andrews & Rothermel 1982). Under P.32/2016 (Pasal 89), aerial suppression is deployed in emergency status or at a Governor's request.

When Start at +0 h (16:00 local). Takes ~1.5 h. Starting at +15 h saves only ~0 ha, because the fire reaches the area first in most runs.

Impact Expected burned area -2 ha (10-90%: 0 to 6 ha); better than no action in 63% of runs. P(B ignites) 54% -> 42%; median arrival +3 h -> +9 h. Lower predicted risk, not a guarantee.

Resources 2 helicopters with 4,000 L buckets for ~1.5 h: 26 drops ~ 104 m³ of water; dip site assumed 5 km away (distance to water not known; set it in Resources), so ~8.6 drops per helicopter-hour. Availability unknown: enter how many helicopters you actually have in Resources.

RECORD

Not sealed. This brief is not a record; seal the forecast in the Record tab before relying on it later.

Research prototype using a simplified fire model. Not firefighting guidance: follow BPBD, Manggala Agni and incident command. Emission numbers are model or satellite estimates, not measurements and not carbon credits.